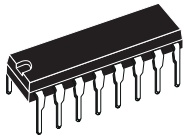
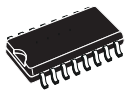
# HCF4050B

## HEX BUFFER/CONVERTER (NON INVERTING)

- PROPAGATION DELAY TIME :
  $t_{PD}$ = 40ns (TYP.) at $V_{DD}$ = 10V $C_L$ = 50pF
- HIGH TO LOW LEVEL LOGIC CONVERSION
- HIGH "SINK" AND "SOURCE" CURRENT CAPABILITY
- QUIESCENT CURRENT SPECIFIED UP TO 20V
- 5V, 10V AND 15V PARAMETRIC RATINGS
- INPUT LEAKAGE CURRENT
  $I_I$ = 100nA (MAX) AT $V_{DD}$ = 18V $T_A$ = 25°C
- 100% TESTED FOR QUIESCENT CURRENT
- MEETS ALL REQUIREMENTS OF JEDEC JESD13B " STANDARD SPECIFICATIONS FOR DESCRIPTION OF B SERIES CMOS DEVICES"



DIP



SOP

### ORDER CODES

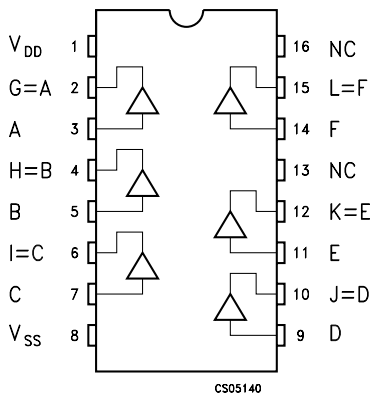
| PACKAGE | TUBE | T & R |
|---|---|---|
| DIP | HCF4050BEY | |
| SOP | HCF4050BM1 | HCF4050M013TR |

### DESCRIPTION

The HCF4050B is a monolithic integrated circuit fabricated in Metal Oxide Semiconductor technology available in DIP and SOP packages.
It is an non inverting Hex Buffer/Converter and feature logic level conversions using only one supply voltage ($V_{DD}$).

The input high level signal ($V_{IH}$) can exceed the $V_{DD}$ supply voltage when these devices are used for logic level conversions. This device is intended for use as CMOS to DTL/TTL converters and can drive directly two DTL/TTL loads ($V_{DD}$=5V, $V_{OL}$≤0.4V and $I_{OL}$≤3.2mA.

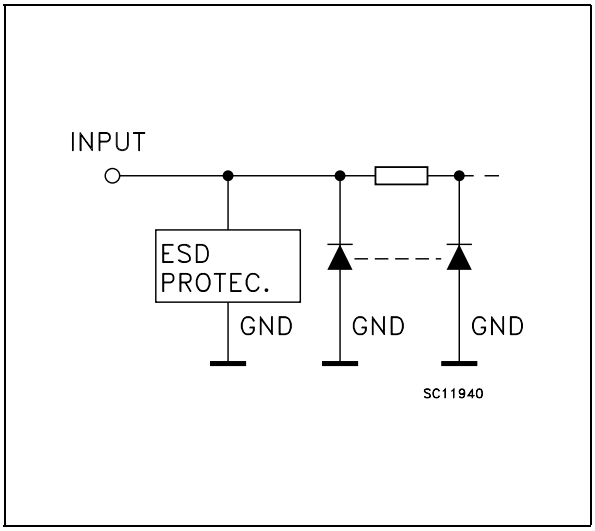
### PIN CONNECTION






This datasheet has been downloaded from http://www.digchip.com at this page

## INPUT EQUIVALENT CIRCUIT




## PIN DESCRIPTION

| PIN No | SYMBOL | NAME AND FUNCTION |
|---|---|---|
| 3, 5, 7, 9, 11, 14 | A, B, C, D, E, F | Data Inputs |
| 2, 4, 6, 10, 12, 15 | G, H, I, J, K, L | Data Outputs |
| 13 , 16 | NC | Not Connected |
| 8 | $V_{SS}$ | Negative Supply Voltage |
| 1 | $V_{DD}$ | Positive Supply Voltage |

## TRUTH TABLE

| INPUTS | OUTPUTS |
|---|---|
| **A, B, C, D,E, F** | **G, H, I, J, K, L** |
| L | L |
| H | H |

## ABSOLUTE MAXIMUM RATINGS

| Symbol | Parameter | Value | Unit |
|---|---|---|---|
| $V_{DD}$ | Supply Voltage | -0.5 to +22 | V |
| $V_I$ | DC Input Voltage | -0.5 to +18 | V |
| $I_I$ | DC Input Current | ± 10 | mA |
| $P_D$ | Power Dissipation per Package | 200 | mW |
| | Power Dissipation per Output Transistor | 100 | mW |
| $T_{op}$ | Operating Temperature | -55 to +125 | °C |
| $T_{stg}$ | Storage Temperature | -65 to +150 | °C |

Absolute Maximum Ratings are those values beyond which damage to the device may occur. Functional operation under these conditions is not implied.
All voltage values are referred to $V_{SS}$ pin voltage.

## RECOMMENDED OPERATING CONDITIONS

| Symbol | Parameter | Value | Unit |
|---|---|---|---|
| $V_{DD}$ | Supply Voltage | 3 to 20 | V |
| $V_I$ | Input Voltage | -0.5 to 15V | V |
| $T_{op}$ | Operating Temperature | -55 to 125 | °C |

## DC SPECIFICATIONS

| Symbol | Parameter | Test Condition | | | | Value | | | | | | | Unit |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | $V_I$ (V) | $V_O$ (V) | $\|I_O\|$ (μA) | $V_{DD}$ (V) | $T_A$ = 25°C | | | -40 to 85°C | | -55 to 125°C | | |
| | | | | | | Min. | Typ. | Max. | Min. | Max. | Min. | Max. | |
| $I_L$ | Quiescent Current | 0/5 | | | 5 | | 0.02 | 1 | | 30 | | 30 | μA |
| | | 0/10 | | | 10 | | 0.02 | 2 | | 60 | | 60 | |
| | | 0/15 | | | 15 | | 0.02 | 4 | | 120 | | 120 | |
| | | 0/20 | | | 20 | | 0.04 | 20 | | 600 | | 600 | |
| $V_{OH}$ | High Level Output Voltage | 0/5 | | <1 | 5 | 4.95 | | | 4.95 | | 4.95 | | V |
| | | 0/10 | | <1 | 10 | 9.95 | | | 9.95 | | 9.95 | | |
| | | 0/15 | | <1 | 15 | 14.95 | | | 14.95 | | 14.95 | | |
| $V_{OL}$ | Low Level Output Voltage | 5/0 | | <1 | 5 | | 0.05 | | | 0.05 | | 0.05 | V |
| | | 10/0 | | <1 | 10 | | 0.05 | | | 0.05 | | 0.05 | |
| | | 15/0 | | <1 | 15 | | 0.05 | | | 0.05 | | 0.05 | |
| $V_{IH}$ | High Level Input Voltage | | 0.5/4.5 | <1 | 5 | 3.5 | | | 3.5 | | 3.5 | | V |
| | | | 1/9 | <1 | 10 | 7 | | | 7 | | 7 | | |
| | | | 1.5/13.5 | <1 | 15 | 11 | | | 11 | | 11 | | |
| $V_{IL}$ | Low Level Input Voltage | | 4.5/0.5 | <1 | 5 | | | 1.5 | | 1.5 | | 1.5 | V |
| | | | 9/1 | <1 | 10 | | | 3 | | 3 | | 3 | |
| | | | 13.5/1.5 | <1 | 15 | | | 4 | | 4 | | 4 | |
| $I_{OH}$ | Output Drive Current | 0/5 | 2.5 | <1 | 5 | -1.25 | -6.4 | | -0.42 | | -0.42 | | mA |
| | | 0/5 | 4.6 | <1 | 5 | -0.51 | -1.6 | | -0.38 | | -0.38 | | |
| | | 0/10 | 9.5 | <1 | 10 | -1.25 | -3.6 | | -1 | | -1 | | |
| | | 0/15 | 13.5 | <1 | 15 | -3.75 | -12 | | -3 | | -3 | | |
| $I_{OL}$ | Output Sink Current | 0/5 | 0.4 | <1 | 5 | 3.2 | 6.4 | | 2.6 | | 2.6 | | mA |
| | | 0/10 | 0.5 | <1 | 10 | 8 | 16 | | 6.6 | | 6.6 | | |
| | | 0/15 | 1.5 | <1 | 15 | 24 | 48 | | 19 | | 19 | | |
| $I_I$ | Input Leakage Current | 0/18 | Any Input | | 18 | | $\pm 10^{-5}$ | ±0.1 | | ±1 | | ±1 | μA |
| $C_I$ | Input Capacitance | | Any Input | | | | 5 | 7.5 | | | | | pF |

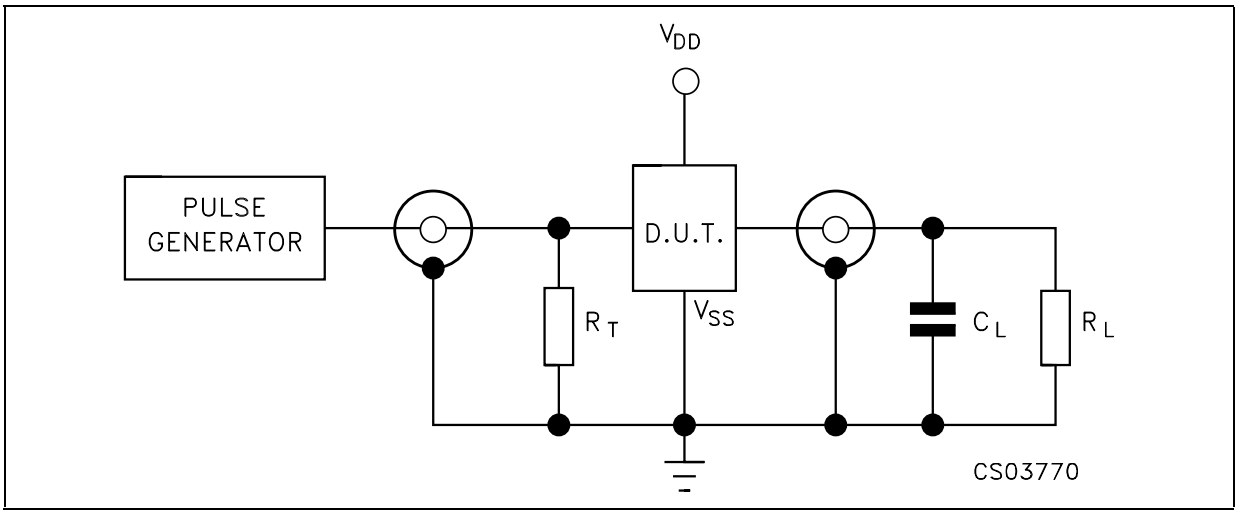
The Noise Margin for both "1" and "0" level is: 1V min. with $V_{DD}$=5V, 2V min. with $V_{DD}$=10V, 2.5V min. with $V_{DD}$=15V

**DYNAMIC ELECTRICAL CHARACTERISTICS** ($T_{amb}$ = 25°C, $C_L$ = 50pF, $R_L$ = 200KΩ, $t_r = t_f$ = 20 ns)

| Symbol | Parameter | Test Condition | | | Value (*) | | | Unit |
|---|---|---|---|---|---|---|---|---|
| | | $V_{DD}$ (V) | $V_I$ (V) | | Min. | Typ. | Max. | |
| $t_{TLH}$ | Output Transition Time | 5 | 5 | | | 80 | 160 | ns |
| | | 10 | 10 | | | 40 | 80 | |
| | | 15 | 15 | | | 30 | 60 | |
| $t_{THL}$ | Output Transition Time | 5 | 5 | | | 30 | 60 | ns |
| | | 10 | 10 | | | 20 | 40 | |
| | | 15 | 15 | | | 15 | 30 | |
| $t_{PLH}$ | Propagation Delay Time | 5 | 5 | | | 70 | 140 | ns |
| | | 10 | 10 | | | 40 | 80 | |
| | | 5 | 10 | | | 45 | 90 | |
| | | 15 | 15 | | | 30 | 60 | |
| | | 5 | 15 | | | 40 | 80 | |
| $t_{PHL}$ | Propagation Delay Time | 5 | 5 | | | 55 | 110 | ns |
| | | 10 | 10 | | | 22 | 55 | |
| | | 5 | 10 | | | 50 | 100 | |
| | | 15 | 15 | | | 15 | 30 | |
| | | 5 | 15 | | | 50 | 100 | |

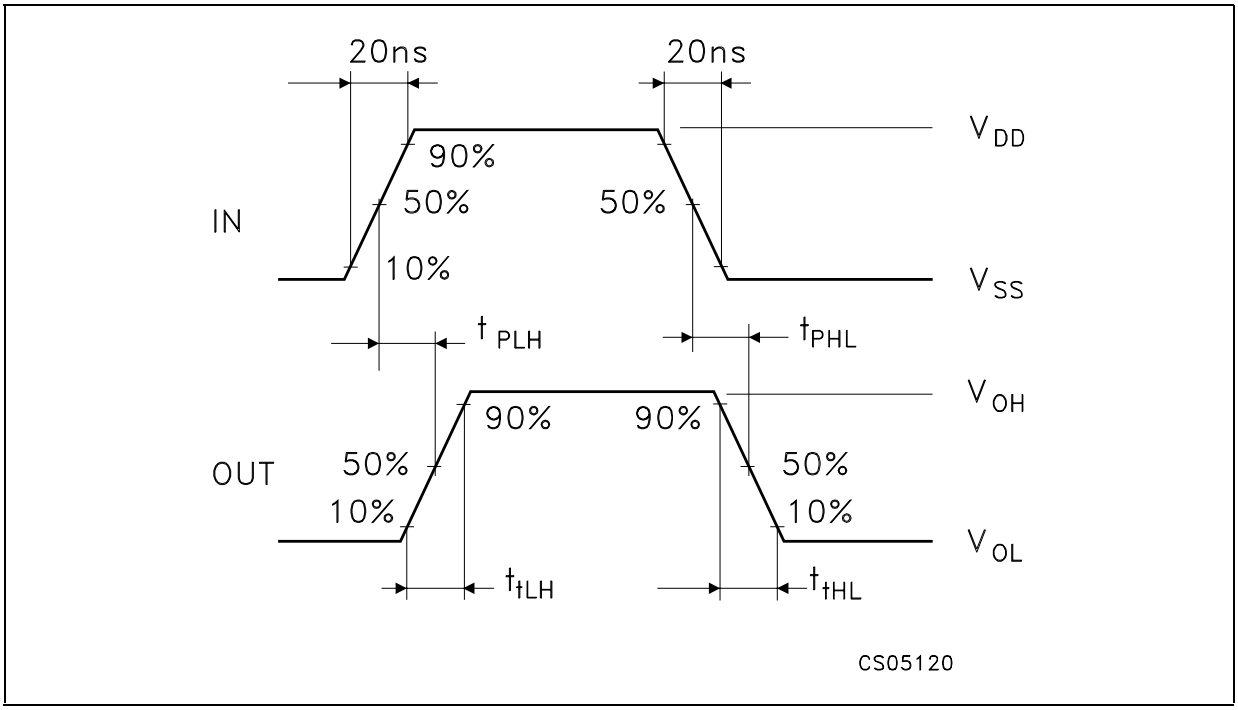
(*) Typical temperature coefficient for all $V_{DD}$ value is 0.3 %/°C.

**TEST CIRCUIT**



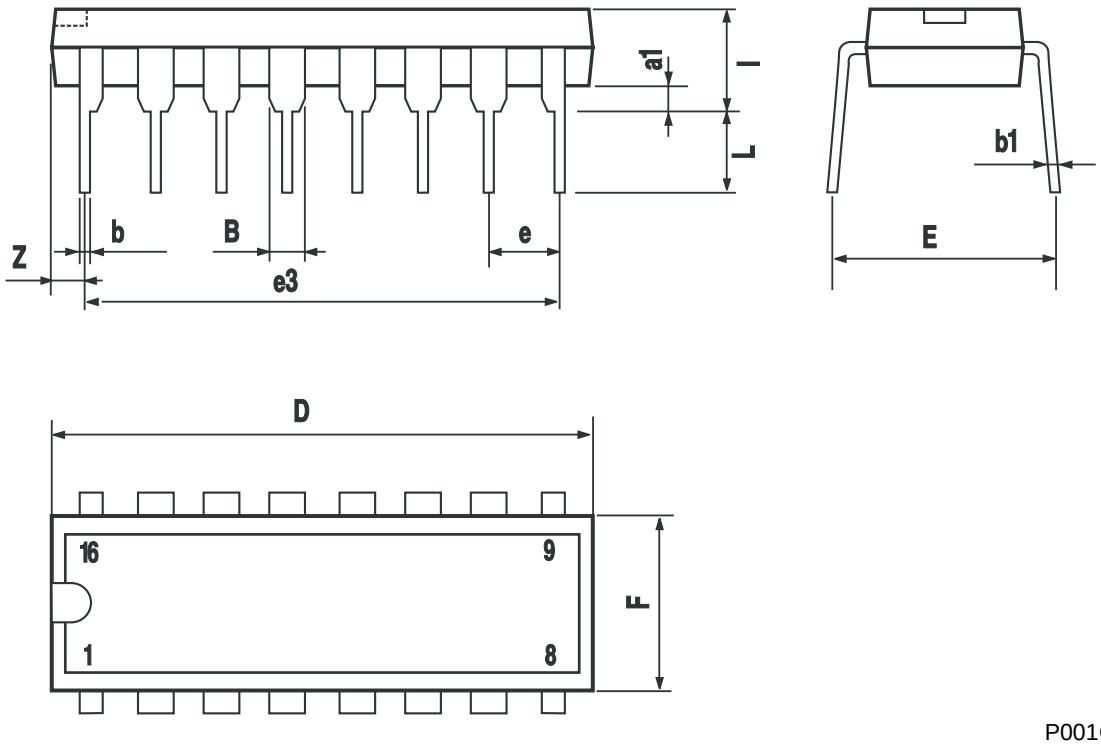

$C_L$ = 50pF or equivalent (includes jig and probe capacitance)
$R_L$ = 200KΩ
$R_T = Z_{OUT}$ of pulse generator (typically 50Ω)

**WAVEFORM : PROPAGATION DELAY TIMES** (f=1MHz; 50% duty cycle)



## Plastic DIP-16 (0.25) MECHANICAL DATA

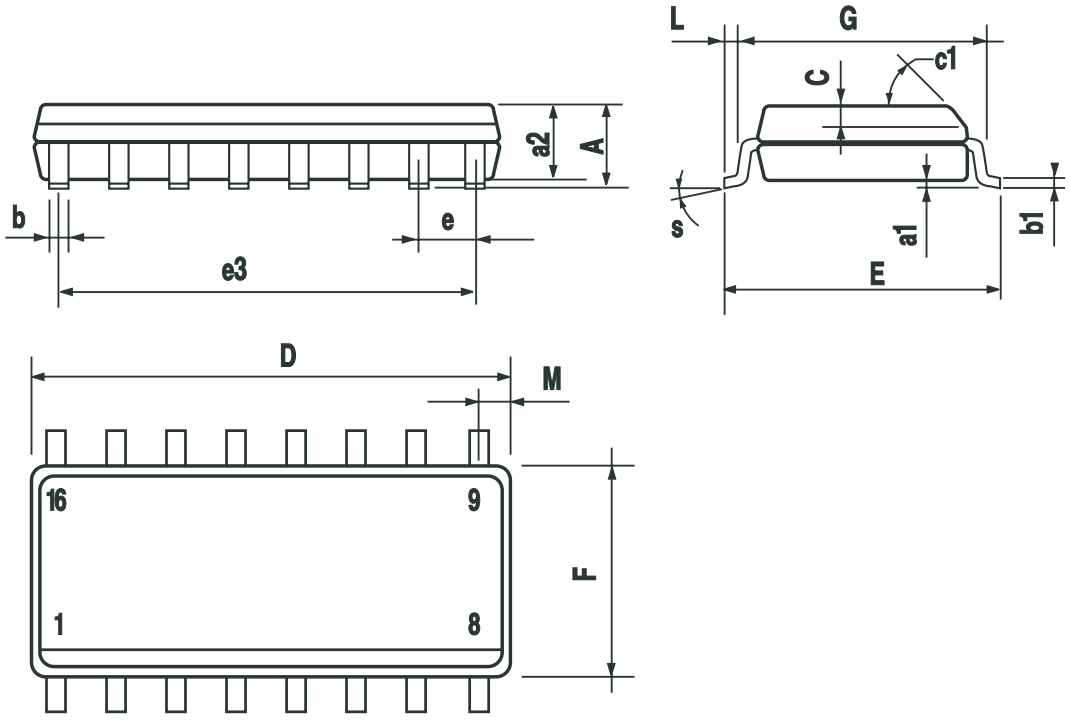
| DIM. | mm. | | | inch | | |
|---|---|---|---|---|---|---|
| | MIN. | TYP | MAX. | MIN. | TYP. | MAX. |
| a1 | 0.51 | | | 0.020 | | |
| B | 0.77 | | 1.65 | 0.030 | | 0.065 |
| b | | 0.5 | | | 0.020 | |
| b1 | | 0.25 | | | 0.010 | |
| D | | | 20 | | | 0.787 |
| E | | 8.5 | | | 0.335 | |
| e | | 2.54 | | | 0.100 | |
| e3 | | 17.78 | | | 0.700 | |
| F | | | 7.1 | | | 0.280 |
| I | | | 5.1 | | | 0.201 |
| L | | 3.3 | | | 0.130 | |
| Z | | | 1.27 | | | 0.050 |




P001C

## SO-16 MECHANICAL DATA

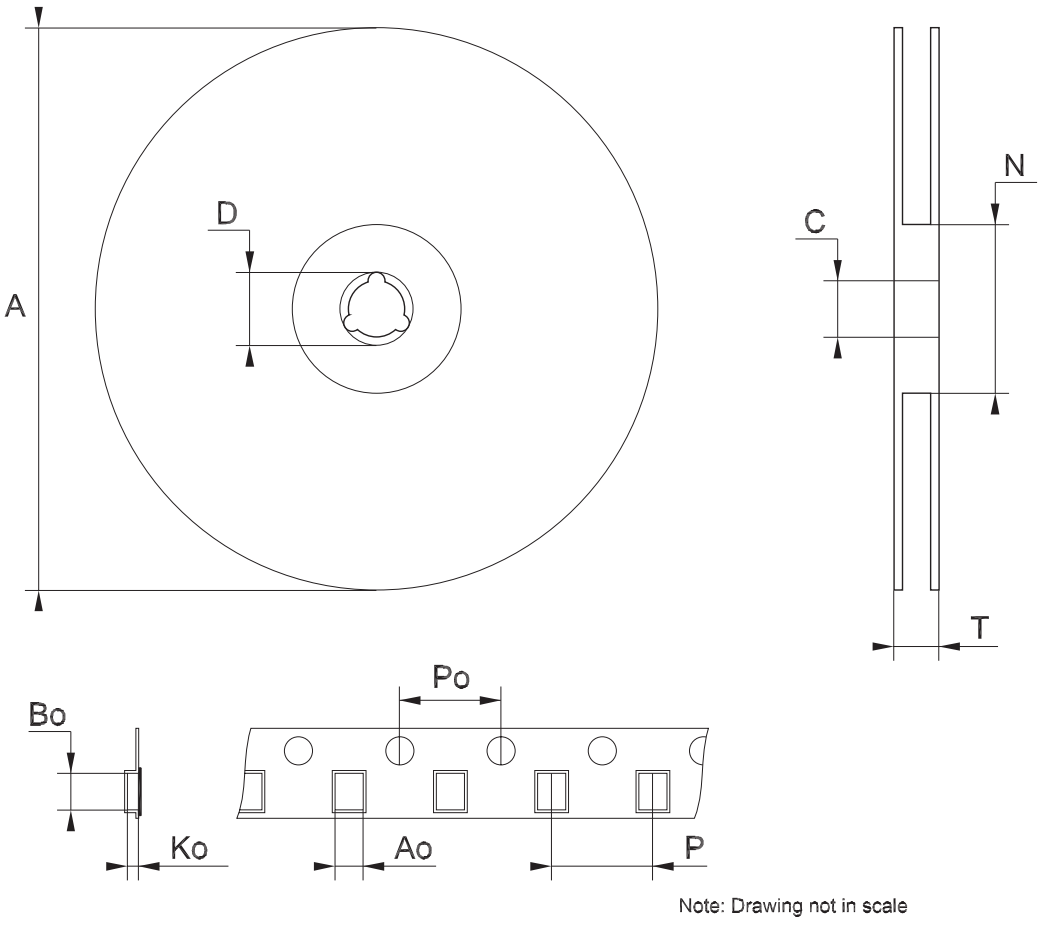
| DIM. | mm. | | | inch | | |
|---|---|---|---|---|---|---|
| | MIN. | TYP | MAX. | MIN. | TYP. | MAX. |
| A | | | 1.75 | | | 0.068 |
| a1 | 0.1 | | 0.2 | 0.004 | | 0.008 |
| a2 | | | 1.65 | | | 0.064 |
| b | 0.35 | | 0.46 | 0.013 | | 0.018 |
| b1 | 0.19 | | 0.25 | 0.007 | | 0.010 |
| C | | 0.5 | | | 0.019 | |
| c1 | 45° (typ.) | | | | | |
| D | 9.8 | | 10 | 0.385 | | 0.393 |
| E | 5.8 | | 6.2 | 0.228 | | 0.244 |
| e | | 1.27 | | | 0.050 | |
| e3 | | 8.89 | | | 0.350 | |
| F | 3.8 | | 4.0 | 0.149 | | 0.157 |
| G | 4.6 | | 5.3 | 0.181 | | 0.208 |
| L | 0.5 | | 1.27 | 0.019 | | 0.050 |
| M | | | 0.62 | | | 0.024 |
| S | 8 | | ° (max.) | | | |




PO13H

## Tape & Reel SO-16 MECHANICAL DATA

| DIM. | mm. | | | inch | | |
|---|---|---|---|---|---|---|
| | MIN. | TYP | MAX. | MIN. | TYP. | MAX. |
| A | | | 330 | | | 12.992 |
| C | 12.8 | | 13.2 | 0.504 | | 0.519 |
| D | 20.2 | | | 0.795 | | |
| N | 60 | | | 2.362 | | |
| T | | | 22.4 | | | 0.882 |
| Ao | 6.45 | | 6.65 | 0.254 | | 0.262 |
| Bo | 10.3 | | 10.5 | 0.406 | | 0.414 |
| Ko | 2.1 | | 2.3 | 0.082 | | 0.090 |
| Po | 3.9 | | 4.1 | 0.153 | | 0.161 |
| P | 7.9 | | 8.1 | 0.311 | | 0.319 |




Note: Drawing not in scale

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